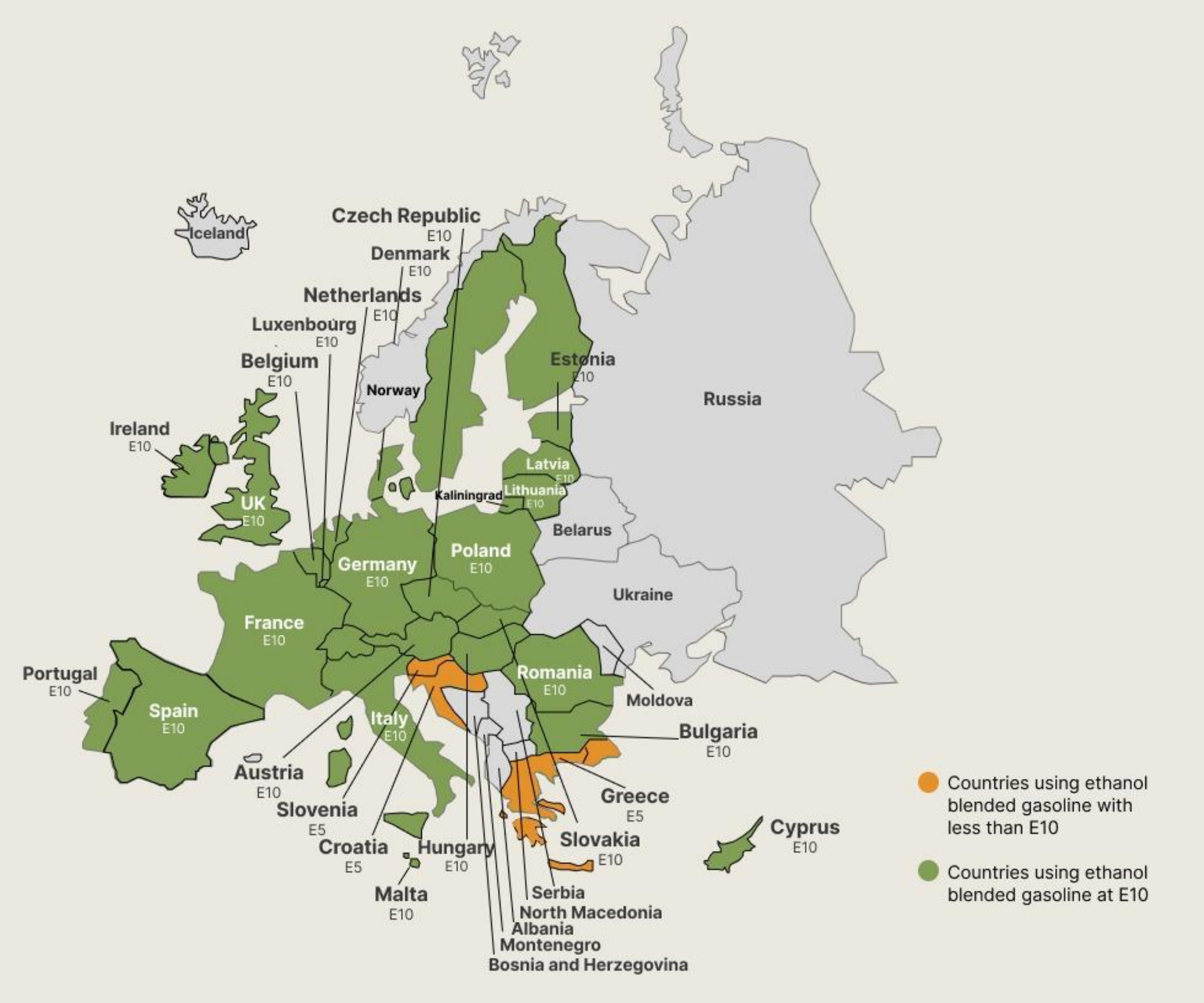
Ethanol Blending in Gasoline - Cyprus

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

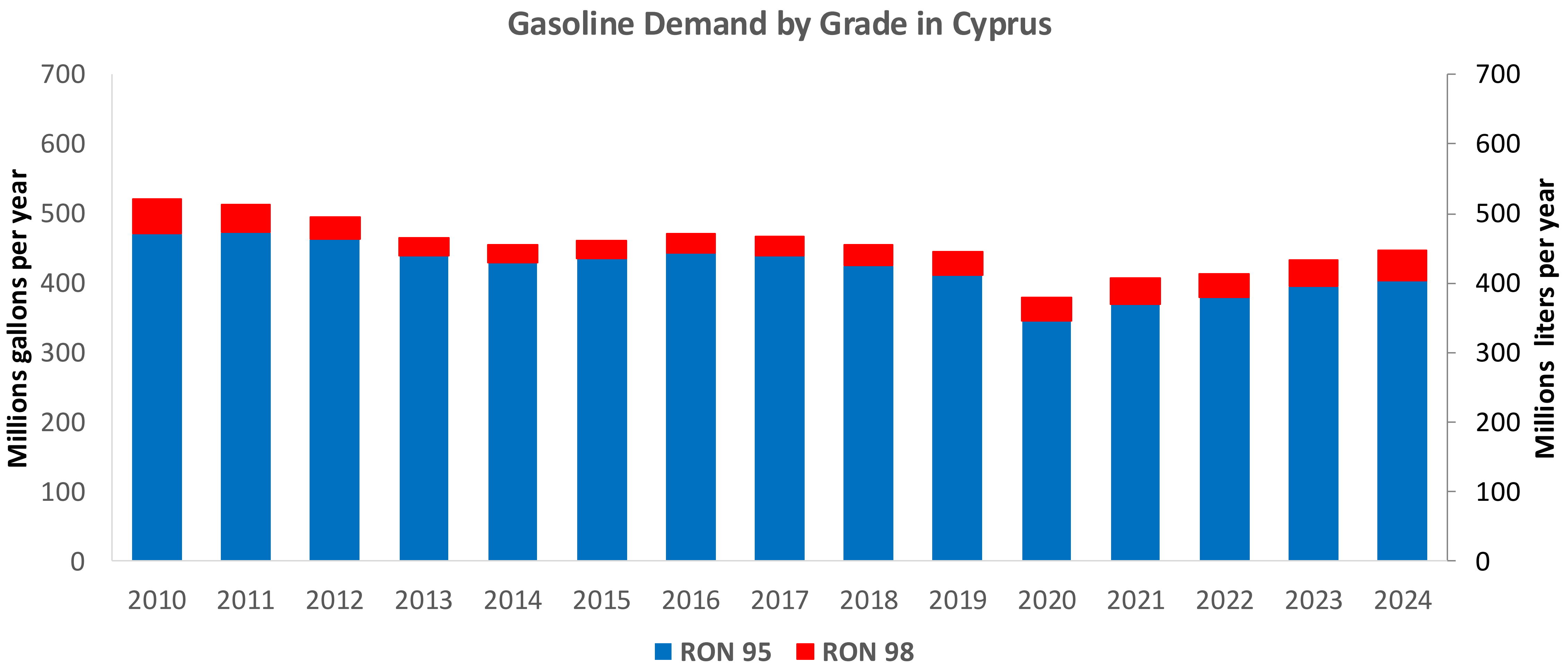
Ethanol Gasoline Blending in Cyprus

In 2024, gasoline consumption in Cyprus was 446 million liters (118 million gallons) and growth rate was 0.1%. Gasoline production and net imports were and 446 million liters (and 118 million gallons) correspondingly. Cyprus complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

Cyprus has no specific ethanol mandate and a target of 14% biofuels in the transport sector by 2030. Ethanol demand was 17.4 million liters in 2024 (4.6 million gallons) equivalent to 3.9% of gasoline demand. Ethanol is supplied by imports.

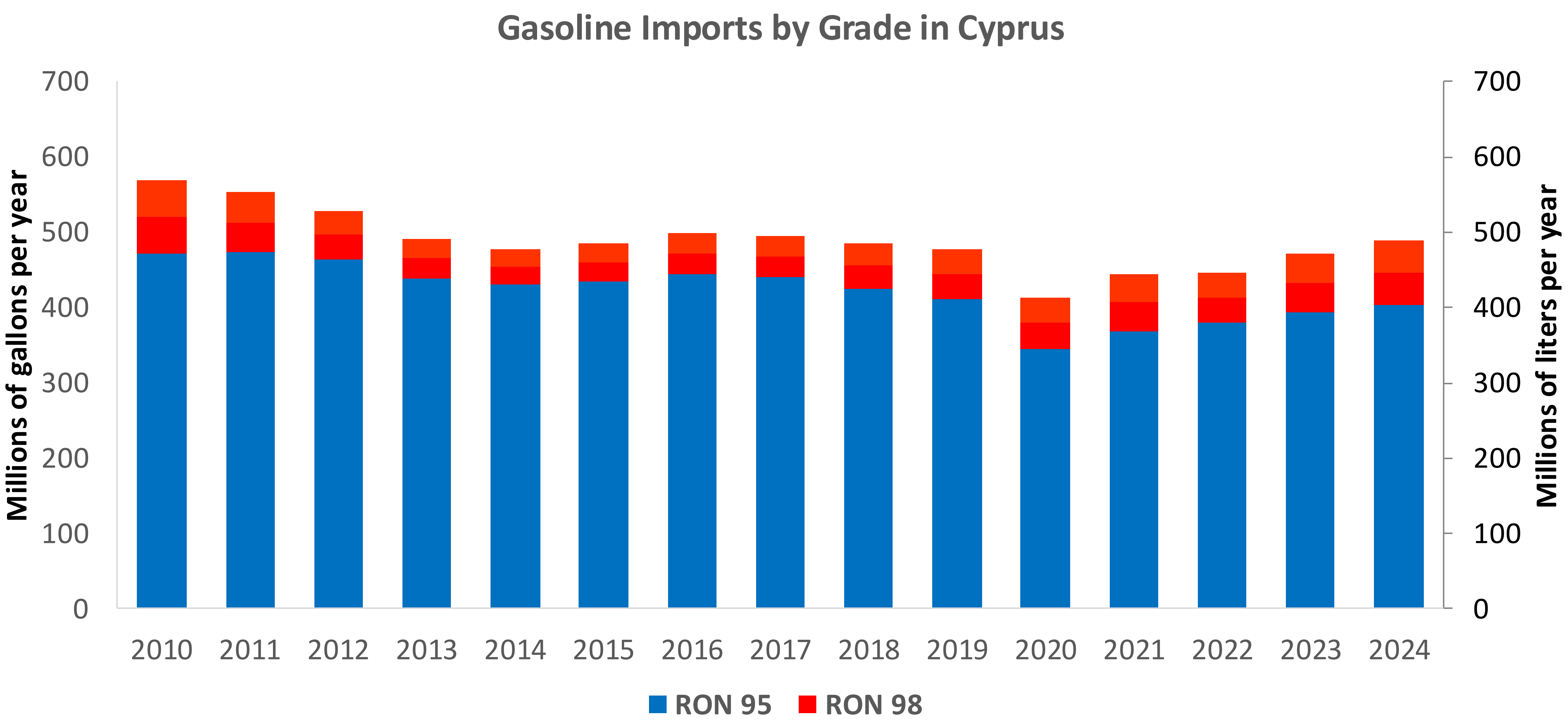
Source: Eurostat, European Commision

Gasoline Demand in Cyprus



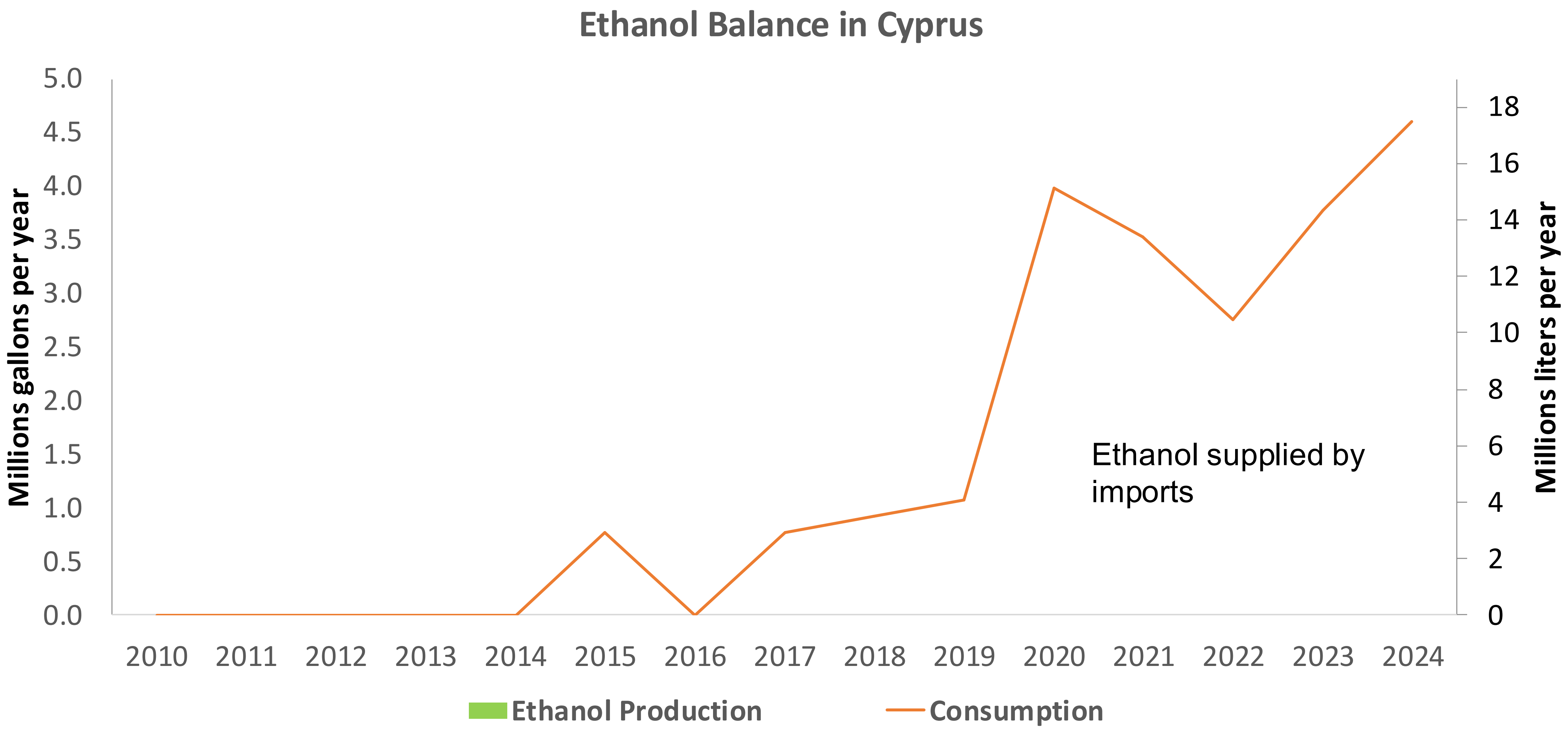
Source: Eurostat

Gasoline Imports in Cyprus



Source: Eurostat

Ethanol Balance in Cyprus



Source: Eurostat, Cyprus Statistics, Global Economy Report

Gasoline Quality in Cyprus

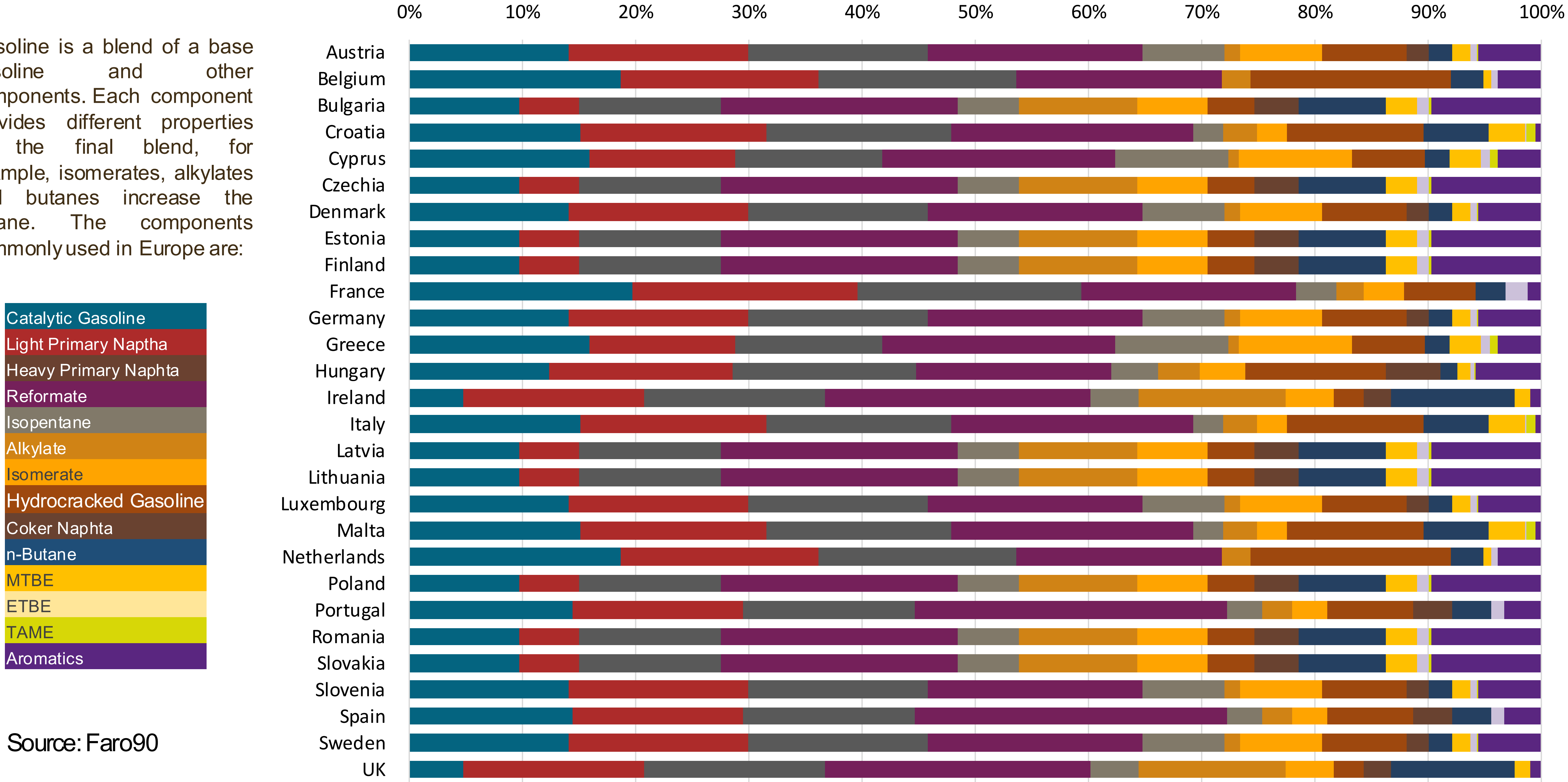
Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:



Source: Faro90

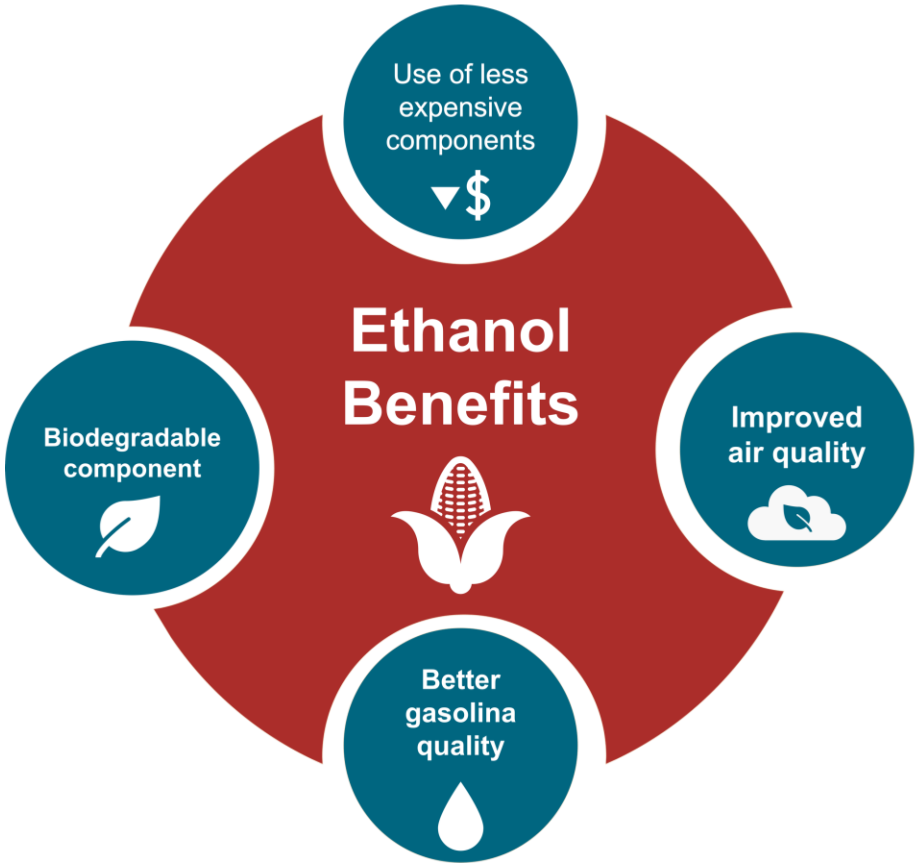
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

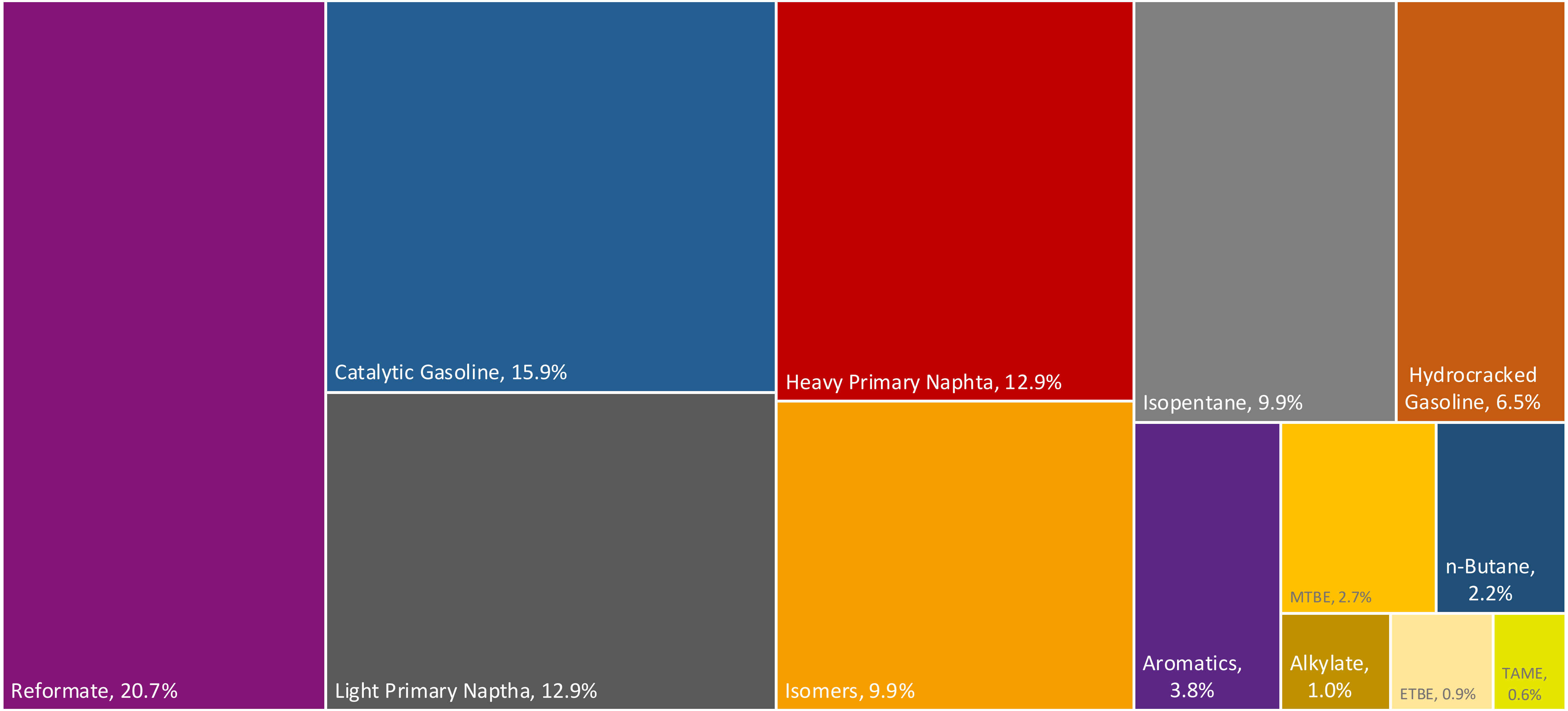
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

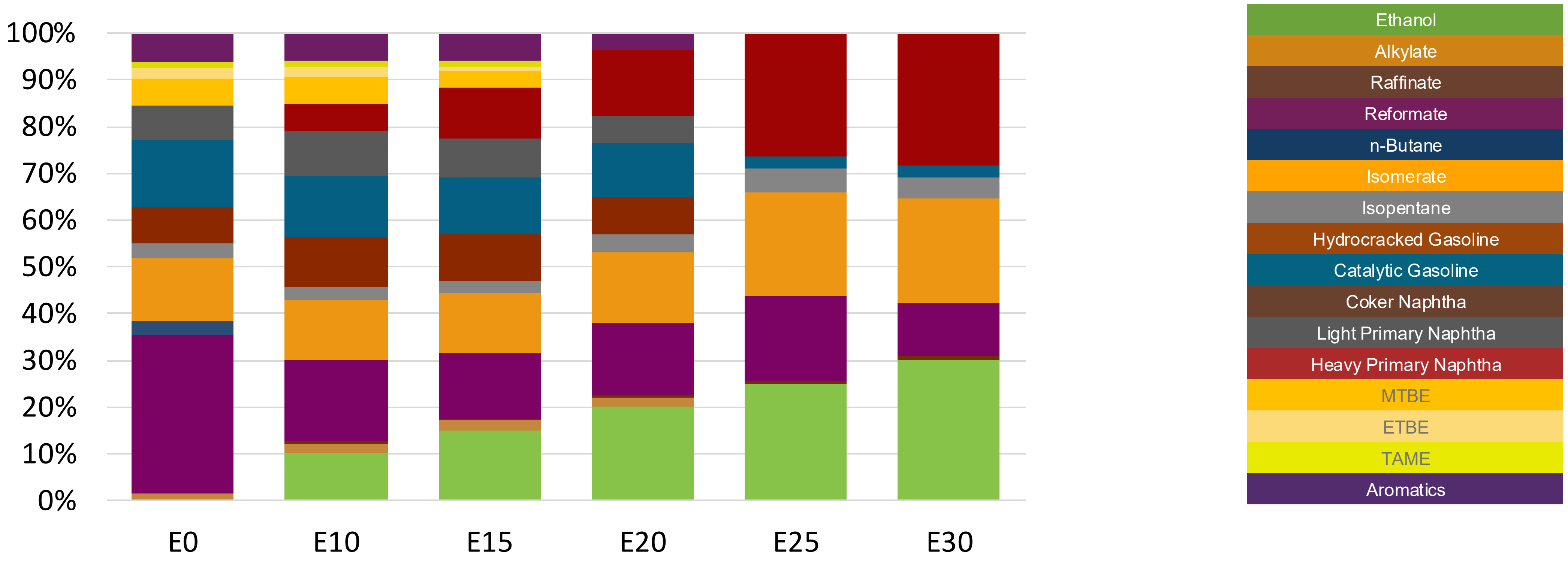


Cyprus Available Blending Components



Source: Faro90

Cyprus – Gasoline 95 – Constant Octane

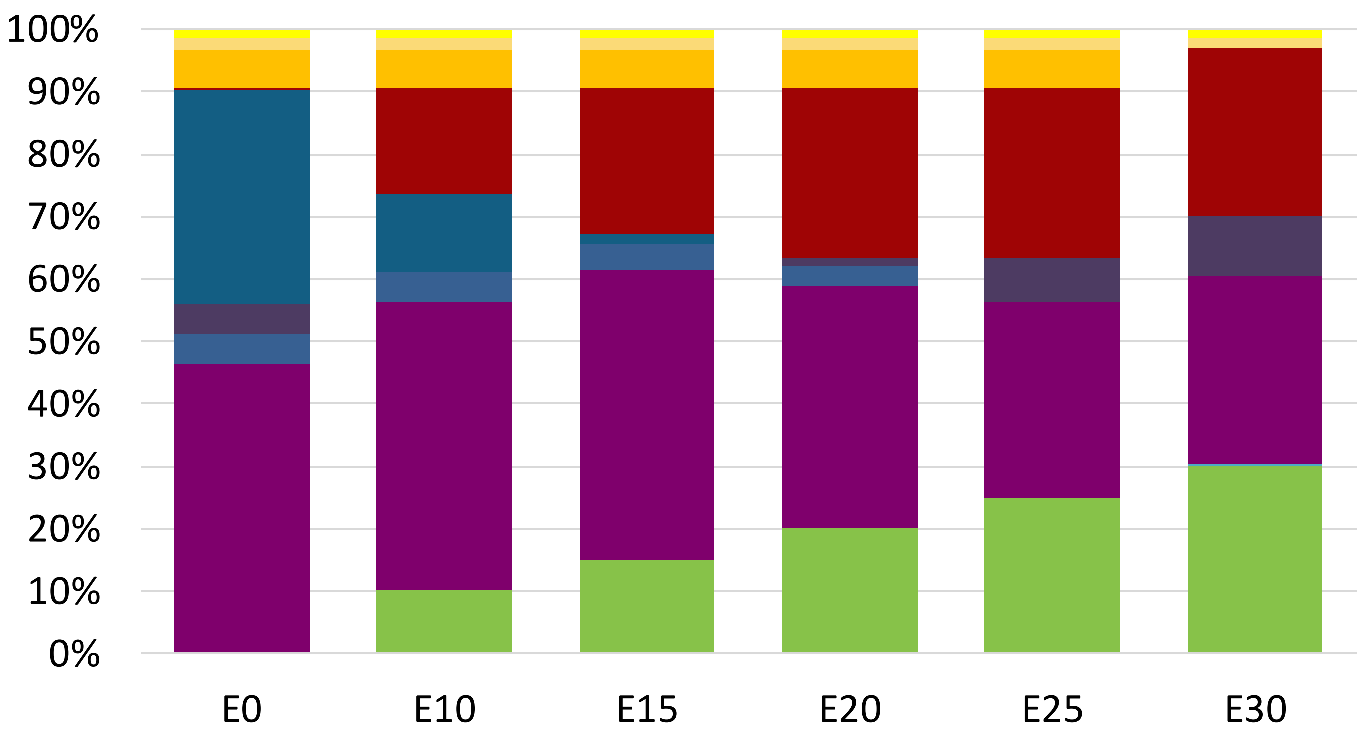


Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US$/gal)	$ 2.258	$ 2.135	$ 2.069	$ 2.002	$ 1.881	$ 1.845

Average CIF 2024 Prices

Source: Faro90

Cyprus - Gasoline 98 - Constant Octane

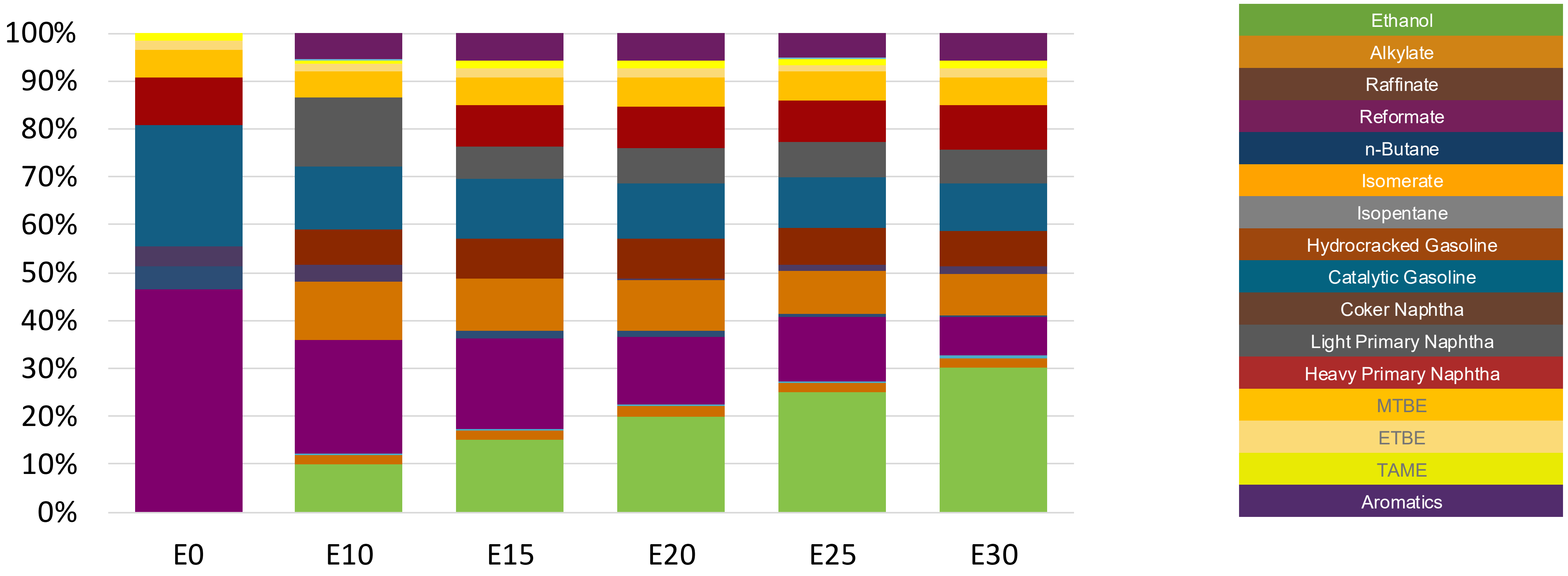


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.330	$ 2.204	$ 2.132	$ 2.057	$ 1.980	$ 1.924

Average CIF 2024 Prices

Source: Faro90

Cyprus – Gasoline 95 – Increased Octane

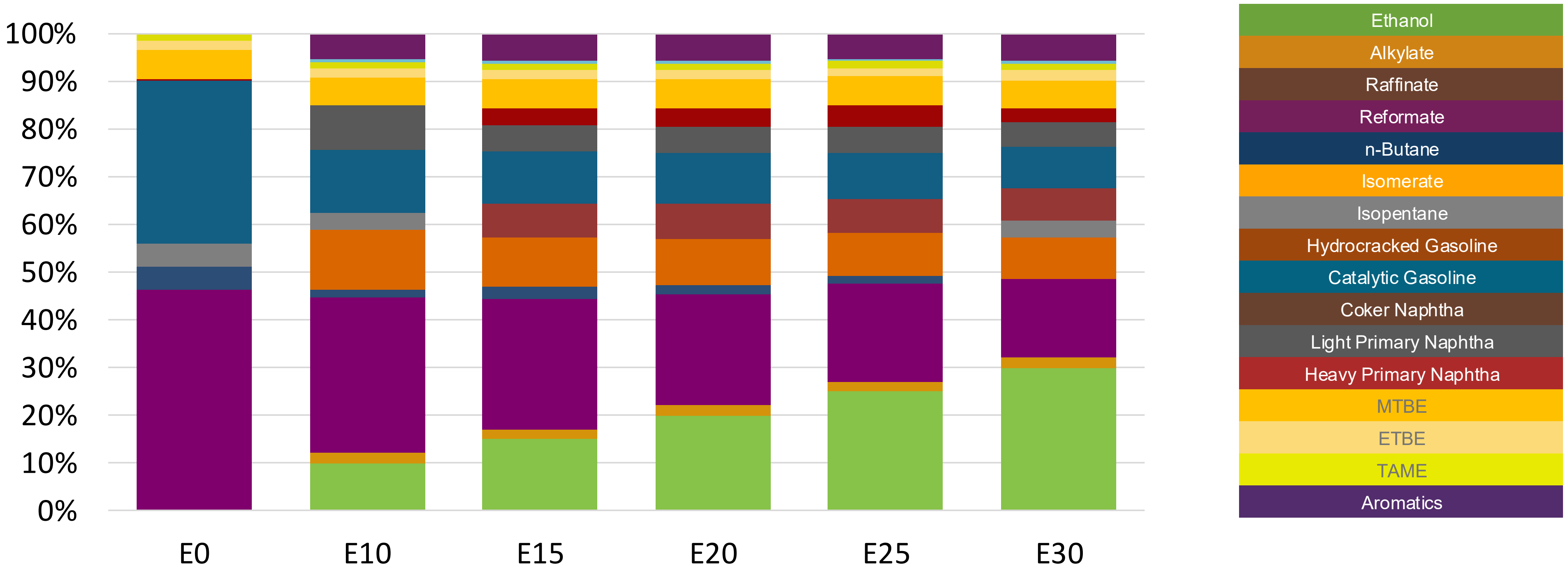


Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US$/gal)	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196	$ 2.196

Average CIF 2024 Prices

Source: Faro90

Cyprus – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US$/gal)	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330	$ 2.330

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

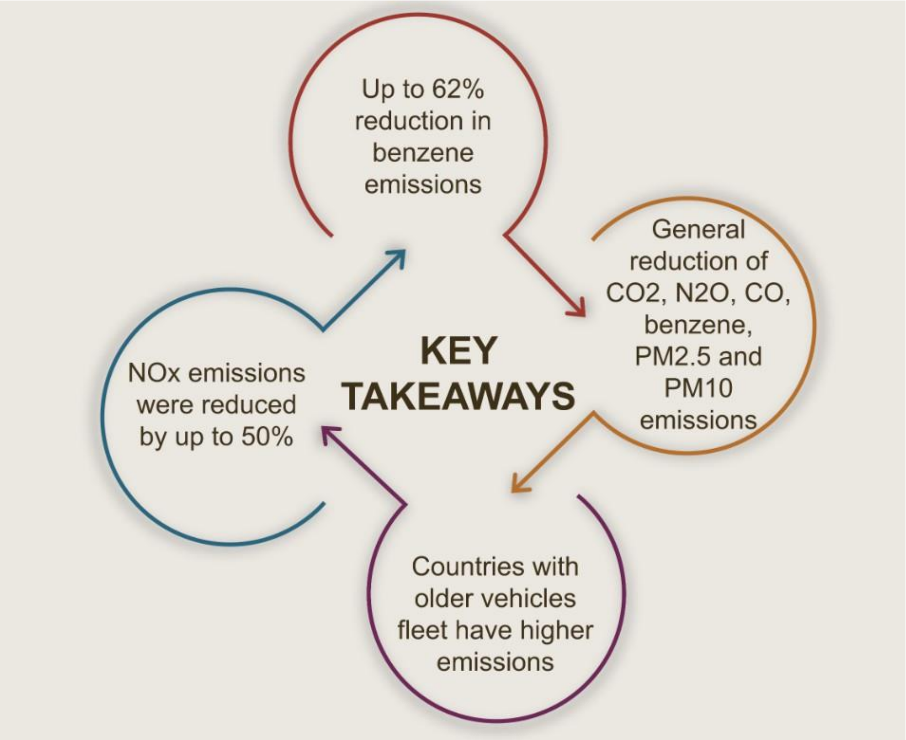
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

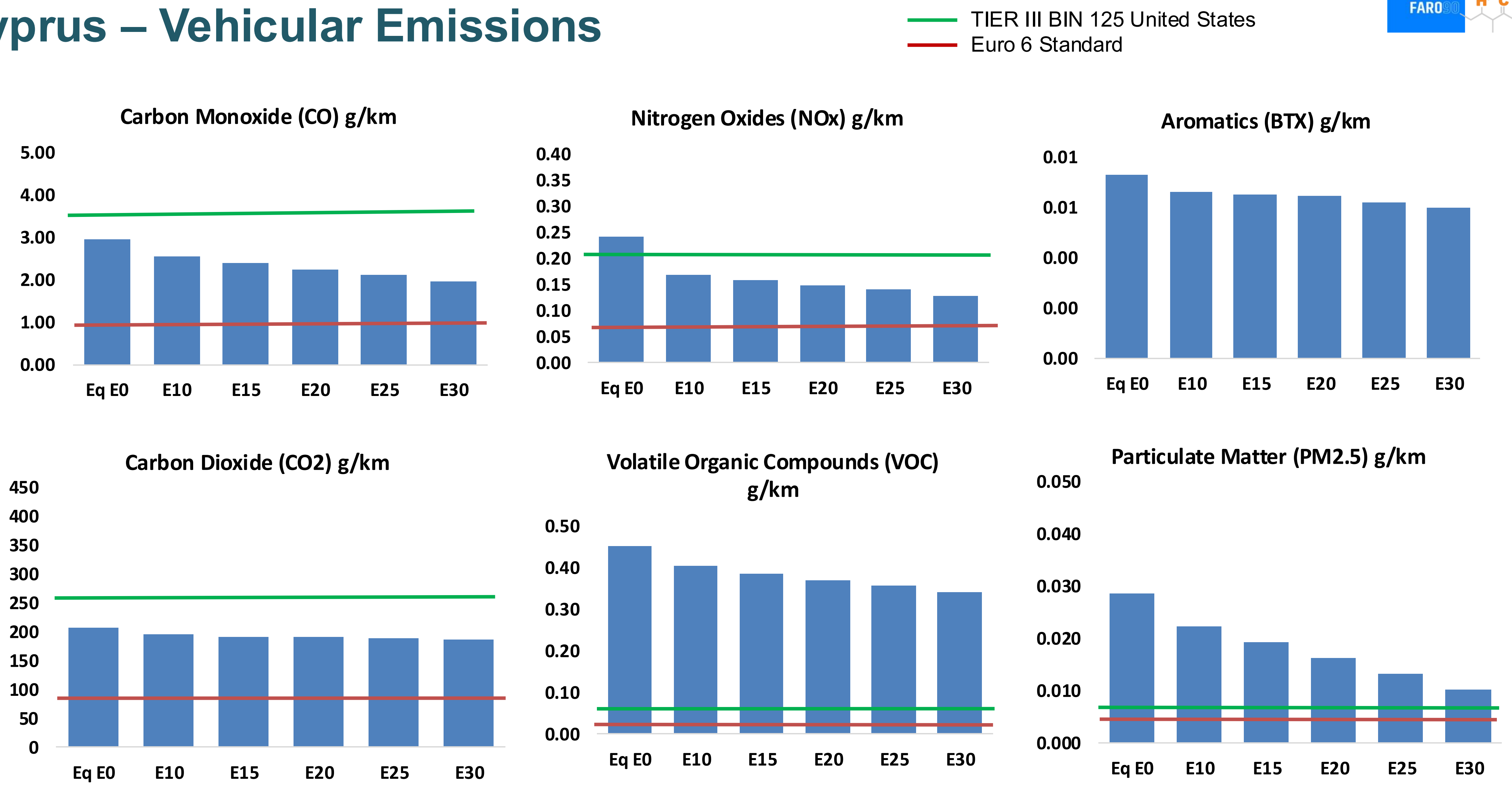
Gasoline Vehicle Fleet - Cyprus

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	556
		Single FI	3-Ways		>8 years	9,050
		Multi FI	3-Ways		>8 years	17,869
			Euro III		>8 years	17,498
			Euro IV		>8 years	87,408
			Euro V		>8 years	47,572
			Euro 6		>8 years	81,464
					4-8 years	144,674
					<4 years	128,128
			Hybrid		All	28,915
		Electric			All	2,691
	Medium	Carburator	3-Ways	PCV Tank	>8 years	1,076
		Single FI	3-Ways		>8 years	14,221
		Multi FI	3-Ways		>8 years	20,318
			Euro III		>8 years	15,556
			Euro IV		>8 years	29,450
			Euro V		>8 years	5,823
			Euro 6		>8 years	7,405
					4-8 years	17,291
					<4 years	15,356
			Hybrid		All	6,847
		Electric			All	637
Sml Engines	Light	4-Cycle	None	PCV	>8 years	27,475
			Euro III		>8 years	17,498
			Euro IV		>8 years	27,516
			Euro V		>8 years	5,212
			Euro 6		>8 years	8,232
					4-8 years	11,453
					<4 years	14,636
			Hybrid		All	3
		Electric			All	1,881
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	545
		Fuel-Injection	3-Ways		>8 years	19,169
			Euro III		>8 years	9,841
			Euro IV		>8 years	18,088
			Euro V		>8 years	3,905
			Euro 6		>8 years	4,082
					4-8 years	7,133
					<4 years	8,392
			Hybrid		All	10,423
		Electric			All	151

Vehicle Fleet: 895,440
Gasoline Vehicle Fleet: 6685,744

Source: Eurostat, ACEA

Cyprus – Vehicular Emissions



Source: Faro90

Cyprus – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	46,762	43,557	40,352	37,802	35,271	33,335	31,171	-14%	-25%
CO2	3,244,976	3,161,534	3,082,728	3,020,749	2,990,079	2,975,062	2,920,223	-5%	-8%
VOC	7,110	6,737	6,364	6,092	5,834	5,640	5,365	-10%	-18%
NOx	3,799	2,849	2,659	2,507	2,364	2,207	2,040	-30%	-38%
SOx	37	34	31	29	27	24	22	-15%	-28%
NH3	1,157	1,146	1,135	1,140	1,153	1,162	1,169	-2%	0%
N2O	43	43	43	43	44	44	45	-1%	2%
CH4	1,608	1,608	1,608	1,632	1,668	1,698	1,727	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	40	38	36	34	33	32	31	-10%	-17%
Acetaldehyde	103	118	173	263	359	410	485	68%	249%
Formaldehyde	372	362	422	496	519	574	625	13%	39%
Benzene	115	110	105	103	102	98	94	-9%	-11%
PM2.5	204	181	158	137	117	95	72	-22%	-43%
PM10	246	218	191	166	141	114	86	-22%	-43%

Source: Faro90

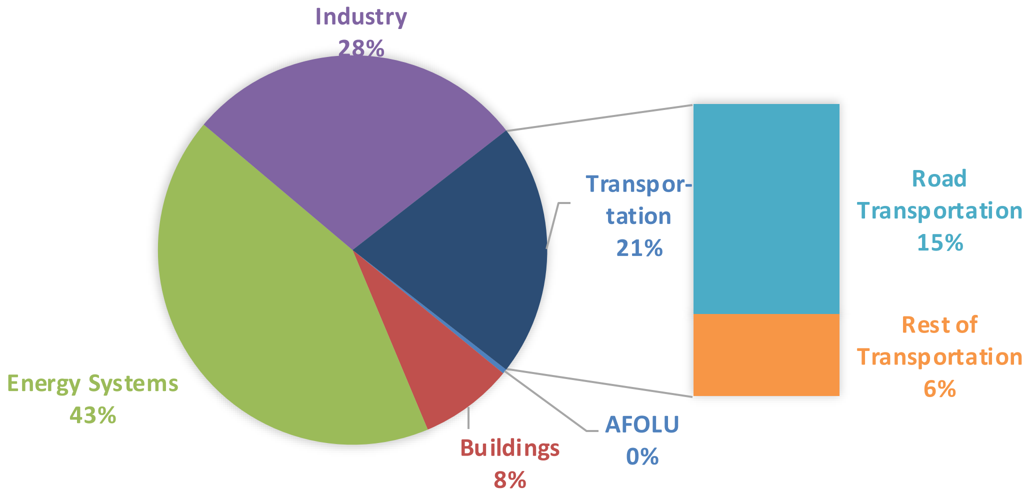
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

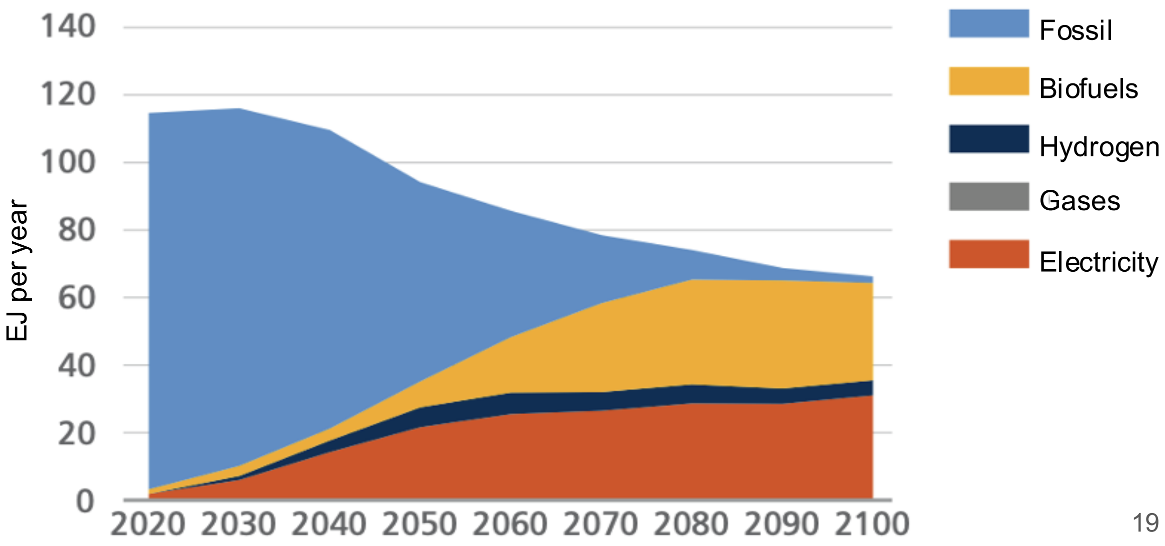
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

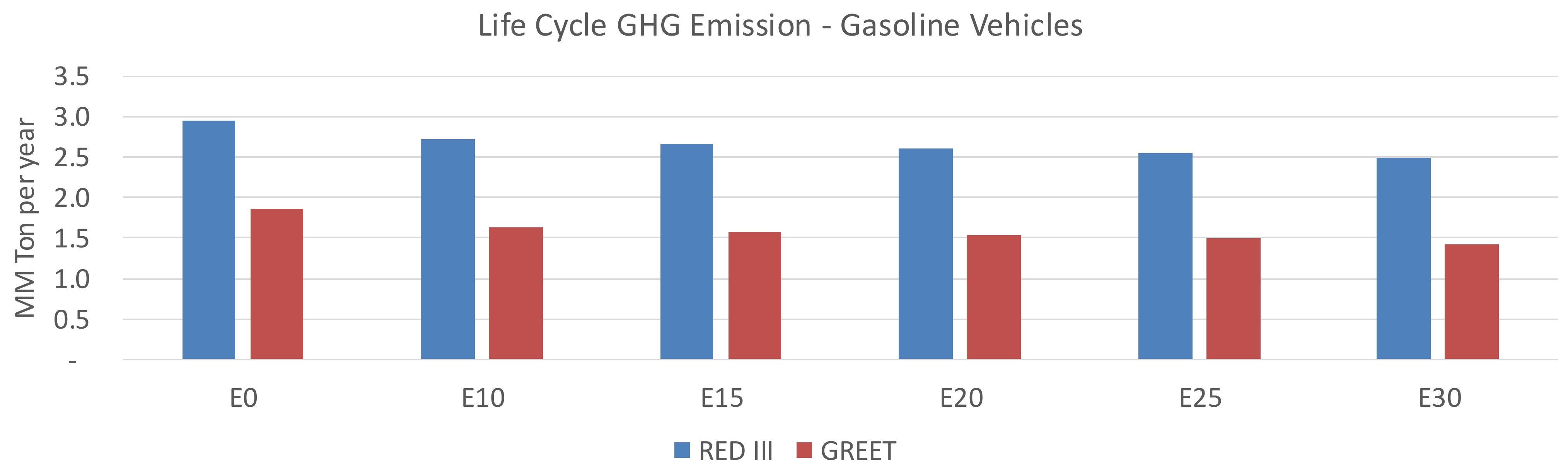
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Cyprus – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	8.3
Target @2035 (MMT/year)	4.7
Delta	3.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.8%	15.0%	17.5%	19.8%	23.0%
RED II/III	0.0%	7.9%	10.1%	11.8%	13.5%	15.6%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	6.1%	7.7%	9.0%	10.2%	11.9%
RED II/III	0.0%	6.5%	8.2%	9.7%	11.0%	12.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90